Lab Exam 2

MTH 308: Numerical Analysis and Scientific Computing–I

Department of Mathematics and Statistics, IIT Kanpur

Time: 5:00PM - 6:00PM, 2026 Even Semester: April 11, 2026

Read the instructions carefully before you start taking the exam:

1. Please write a single MATLAB script file named `LastnameRollNumber.m` and publish it as a PDF file (click on the `PUBLISH` tab → click the down arrow on the `Publish` button → select `Edit Publishing Options..` → set the `output file format` to `PDF` → click `Publish`). For example, my submitted files should be `Biswas260102.m` and `Biswas260102.pdf`.
2. Your solution should be adequately commented to clearly explain your logic.
3. **You must complete the coding, generate both files, and save them in a folder on the Desktop by 5:50 PM**.
4. Access to the Hello IITK portal (via the bookmark `"Log in | HelloIITK"` in the browser) will be provided during **5:50 PM–6:00 PM**. Submission of *both* files must be completed *only* within this time window. **Failure to submit *both* files will result in the solution not being graded, and no second chance will be given**.
5. Any form of dishonest practice during the lab exam will result in failure of the exam.
6. **Mobile phones, smart watches, calculators, and other electronic devices are prohibited**. Possession will result in disqualification. Keep all devices switched off outside the lab.

Problem [10]

Use Newton's method to find a solution to the nonlinear system $F(x) = 0$, where $F : \mathbb{R}^3 \to \mathbb{R}^3$ is defined by

$$F(x) = \begin{bmatrix} x_1^3 + x_1^2 x_2 - x_1 x_3 + 6 \\ e^{x_1} + e^{x_2} - x_3 \\ x_2^2 - 2x_1 x_3 - 4 \end{bmatrix}.$$

Iterate until $\|x^{(k)} - x^{(k-1)}\|_\infty < 10^{-8}$ starting from $x^{(0)} = (-1, -2, 1)^T$. Here $\|\cdot\|_\infty$ denotes the infinity norm.

1. Print the iteration history (starting from $k = 1$) at each iteration in the format

$$k,\ x_1^{(k)},\ x_2^{(k)},\ x_3^{(k)},\ \|x^{(k)} - x^{(k-1)}\|_\infty,\ \|F(x^{(k)})\|_\infty.$$

2. In MATLAB, you may use the following commands for formatted output.

   Place this before the iteration loop:

```
fprintf('k\t   x1\t\t   x2\t\t   x3\t\t   ||dx||_inf\t   ||F(x)||_inf\n');
```

   To print the iteration history at each step, place the following inside the loop:

```
fprintf('%d\t% .6f\t% .6f\t% .6f\t% .2e\t% .2e\n', ...
        k, x(1), x(2), x(3), norm(x - x_old, inf), norm(Fx, inf));
```

   where `x_old` stores $x^{(k-1)}$, x stores $x^{(k)}$, and `Fx = F(x)`.

3. Print the final approximate solution x using a similar MATLAB formatting:

```
fprintf('x = [% .6f, % .6f, % .6f]^T\n', x(1), x(2), x(3));
```